RESEARCH IN SCIENCE & TECHNOLOGICAL EDUCATION

01/03

NEW ARTICLE

Could AI-agents improve college students' scientific argumentation ability? A case study in ecology course

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DOI **doi.org/10.1080/02635143.2026.2753076**

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WHAT DID THE RESEARCHERS INVESTIGATE?

- Scientific argumentation is hard to improve in traditional classes because of individual differences.
- The authors built an AI agent based on the CER+R framework to support personalised, data-driven practice.
- Eighteen Ecology students in eastern China used the agent either before or after practising independently in a crossover design.

Liu et al. (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2753076

WHAT DID THEY FIND?

- Students who practised independently before using the agent showed more interactive and constructive engagement.
- Among the 12 students with complete test data, argumentation performance improved significantly from pre-test to mid-test and post-test.
- Rebuttals often lacked later evidence or reasoning, so the authors suggest additional scaffolding is needed.

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